Nonparametric Statistics

Chapter 2: Density Estimation

Projection Density Estimator

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Outline

1. Orthonormal Bases in $L^2([0,1])$
2. Construction of the Projection Estimator
3. Risk Analysis
4. Optimal Convergence Rate
5. Adaptive Estimation and Model Selection
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Setting the Stage

Problem Setup

Given i.i.d. observations $X_1, \ldots, X_n \sim f$ on $[0,1]$, we want to estimate the density $f \in L^2([0,1])$ by **projecting onto a finite-dimensional subspace** S_D.

Key idea: Represent f as an infinite series

$$f = \sum_{j=1}^{\infty} a_j \varphi_j$$

in an orthonormal basis $\{\varphi_j\}$, then **estimate only the first** D **coefficients** from data.

The integer D plays the same role as the **bandwidth** h in kernel estimation:
it is the **smoothing parameter** controlling the bias–variance tradeoff.

Orthogonal Projection in $L^2([0,1])$

Let $S_D \subset L^2([0,1])$ be a D-dimensional subspace with orthonormal basis $(\varphi_1, \ldots, \varphi_D)$:

$$\langle \varphi_j, \varphi_k \rangle = \int_0^1 \varphi_j(x)\varphi_k(x)\,dx = \delta_{jk}.$$

The **orthogonal projection** of $g \in L^2([0,1])$ onto S_D is $g_D = P_{S_D}(g)$, characterized by:

Orthogonal Projection

$$g_D = \sum_{j=1}^{D} \alpha_j(g)\,\varphi_j, \qquad \alpha_j(g) = \langle g, \varphi_j \rangle = \int_0^1 g(x)\varphi_j(x)\,dx.$$

Equivalently: $g_D \in S_D$ and $g - g_D \perp S_D$, i.e. $\langle g - g_D, \varphi_j \rangle = 0$ for all j.

Parseval's identity: For $h = \sum_{j=1}^{D} \alpha_j \varphi_j \in S_D$, $\quad \|h\|^2 = \sum_{j=1}^{D} \alpha_j^2$.

Trigonometric Basis

Definition — Trigonometric Basis

For $D = 2d + 1$ (odd), define on $[0, 1]$:

$$\varphi_1(x) = 1, \quad \varphi_{2j}(x) = \sqrt{2}\cos(2\pi jx), \quad \varphi_{2j+1}(x) = \sqrt{2}\sin(2\pi jx), \quad j = 1, \ldots, d.$$

Key properties:

- $(\varphi_1, \varphi_2, \ldots, \varphi_D)$ is an orthonormal basis of S_D.
- The spaces are **nested**: $S_1 \subset S_2 \subset S_3 \subset \cdots$
- Adding a new term does not modify existing coefficients.
- The basis functions are **smooth**: good for estimating smooth densities.

Proposition (Norm bound)

For any $h = \sum_{j=1}^{D} \alpha_j \varphi_j \in S_D$:

$$\|h\|_\infty \leq \sqrt{D}\,\|h\|, \qquad \left\| \sum_{j=1}^{D} \varphi_j^2 \right\|_\infty = D.$$

Uniform Histogram Basis

Definition — Histogram Basis

Partition $[0,1]$ into D equal bins. Define:

$$\varphi_j(x) = \sqrt{D}\, 1_{\left[\frac{j-1}{D}, \frac{j}{D}\right)}(x), \qquad j = 1, \ldots, D.$$

Key properties:

- Orthonormal: $\varphi_j(x)\varphi_k(x) = 0$ for $j \neq k$ (disjoint supports).
- $\int \varphi_j^2 = 1$ (normalizing factor $\sqrt{D}$).
- Spaces are NOT nested: changing $D \to D+1$ redefines all basis functions.
- **Simple to compute**, but gives discontinuous (step-function) estimates.

Connection to histogram: The projection estimator onto the histogram basis is exactly the classical histogram density estimator with D equal bins.

Piecewise Polynomial Basis (Legendre)

Definition

For integer d and polynomial order r, define:

$$\varphi_{j,k}^{(d)}(x) = \sqrt{2d}\, Q_k(2dx - 2j + 1)\, 1_{\left[\frac{j-1}{d}, \frac{j}{d}\right)}(x),$$

where $(Q_k)_{k\geq 0}$ is the **Legendre basis** of $L^2([-1,1])$. The resulting space S_D has dimension $D = d(r+1)$.

Legendre polynomials Q_k:

$$Q_0 = \tfrac{1}{\sqrt{2}}, \quad Q_1(x) = \sqrt{\tfrac{3}{2}}\, x, \quad Q_2(x) = \tfrac{1}{2}\sqrt{\tfrac{5}{2}}(3x^2 - 1), \quad \dots$$

Proposition 2.1.5

For any $h \in S_D$: $\quad \|h\|_\infty \leq \Phi_0 \sqrt{D}\, \|h\|, \quad$ with $\Phi_0 = \sqrt{r+1}$.

Comparing the Three Bases

Property	**Trigonometric**	**Histogram**	**Piecewise Poly.**
Smoothness of basis	Very high	None	Moderate
Nested spaces	Yes	No	No
Best for	Smooth f	Step-like f	General f
$\|h\|_\infty \leq$	$\sqrt{D}\|h\|$	$\sqrt{D}\|h\|$	$\Phi_0\sqrt{D}\|h\|$
ϕ_0 constant	1	1	$r+1$

General result: For all classical bases (trigonometric, histogram, splines, ...)

$$\left\| \sum_{j=1}^{D} \varphi_j^2 \right\|_\infty \leq \phi_0 D$$

for some constant ϕ_0 depending only on the basis. This is the **key property** used in the variance bound.

Extending to an Arbitrary Interval $[a, b]$

All bases above are defined on $[0, 1]$. In practice, data lives on some interval $[a, b]$.

Change of Variables

Given an orthonormal basis (φ_j) of $L^2([0, 1])$, define:

$$\tilde{\varphi}_j(x) = \frac{1}{\sqrt{b-a}} \varphi_j\left(\frac{x-a}{b-a}\right).$$

Then $(\tilde{\varphi}_j)$ is an orthonormal basis of $L^2([a, b])$:

$$\int_a^b \tilde{\varphi}_j(u)\tilde{\varphi}_k(u)\,du = \int_0^1 \varphi_j(x)\varphi_k(x)\,dx = \delta_{jk}.$$

In practice: Transform data $X_i \mapsto U_i = (X_i - a)/(b-a) \in [0, 1]$, apply the projection estimator on $[0, 1]$, then rescale.

From Fourier Coefficients to Estimator

The density $f \in L^2([0,1])$ expands as:

$$f = \sum_{j=1}^{\infty} a_j \varphi_j, \qquad a_j = \langle \varphi_j, f \rangle = \int_0^1 \varphi_j(x) f(x)\, dx = \mathbb{E}[\varphi_j(X_1)].$$

Key observation: The Fourier coefficient a_j is the **expected value of** $\varphi_j(X_1)$.

By the Law of Large Numbers, the natural estimator is:

$$\hat{a}_j = \frac{1}{n} \sum_{i=1}^{n} \varphi_j(X_i) \quad \Rightarrow \quad \mathbb{E}[\hat{a}_j] = a_j \text{ (unbiased)}.$$

The **best D-dimensional approximation** (oracle) of f is:

$$f_D = \sum_{j=1}^{D} a_j \varphi_j.$$

We cannot compute f_D since a_j depends on the unknown f.
$\hookrightarrow$ Replace a_j by $\hat{a}_j$.

The Projection Estimator

Definition 2.2.1 — Projection Estimator

Given an orthonormal basis $(\varphi_1, \ldots, \varphi_D)$ and i.i.d. sample $X_1, \ldots, X_n \sim f$:

$$\hat{f}_D = \sum_{j=1}^{D} \hat{a}_j \, \varphi_j, \qquad \hat{a}_j = \frac{1}{n} \sum_{i=1}^{n} \varphi_j(X_i).$$

Remarks:

- $\hat{f}_D$ is a **linear estimator** in the observations X_i.
- Each $\hat{a}_j$ is unbiased: $\mathbb{E}[\hat{a}_j] = a_j$.
- D is the **smoothing parameter** to be chosen.

Limitation: $\hat{f}_D$ may take negative values (unlike KDE), so it is not necessarily a valid density function.

Alternative Derivation: Contrast Minimization

An equivalent way to derive $\hat{f}_D$ is via **contrast minimization**.

Goal: minimize $\|f - h\|^2$ over $h \in S_D$. Since $\|f - h\|^2 = \|h\|^2 - 2\langle h, f\rangle + \|f\|^2$ and $\|f\|^2$ is fixed, this is equivalent to minimizing:

Contrast Function

$$\gamma_n(h) = \|h\|^2 - \frac{2}{n}\sum_{i=1}^{n} h(X_i).$$

The minimizer: Setting $\partial\gamma_n/\partial a_{j_0} = 0$ gives:

$$\tilde{a}_j = \frac{1}{n}\sum_{i=1}^{n} \varphi_j(X_i) = \hat{a}_j.$$

Conclusion: Both approaches (Fourier coefficient estimation and contrast minimization) yield the **same estimator** $\tilde{f}_D = \hat{f}_D$.

Bias–Variance Decomposition

The quadratic risk of $\hat{f}_D$ is $\mathbb{E}[\|f - \hat{f}_D\|^2]$.

Since $f - f_D \perp S_D$ and $f_D - \hat{f}_D \in S_D$, by the **Pythagorean theorem**:

$$\|f - \hat{f}_D\|^2 = \|(f - f_D) + (f_D - \hat{f}_D)\|^2 = \|f - f_D\|^2 + \|f_D - \hat{f}_D\|^2.$$

Bias–Variance Decomposition

$$\mathbb{E}\left[\|f - \hat{f}_D\|^2\right] = \underbrace{\|f - f_D\|^2}_{\text{Bias}^2 \text{ (approximation error)}} + \underbrace{\mathbb{E}\left[\|f_D - \hat{f}_D\|^2\right]}_{\text{Variance (estimation error)}}.$$

- **Bias**^{2} $= \|f - f_D\|^2$: how well S_D approximates f. Decreases as $D \uparrow$.
- **Variance** $= \mathbb{E}[\|f_D - \hat{f}_D\|^2]$: statistical fluctuation from estimating a_j. Increases as $D \uparrow$.

Computing the Variance Term

Using orthonormality of (φ_j):

$$\|f_D - \hat{f}_D\|^2 = \left\| \sum_{j=1}^{D} (\hat{a}_j - a_j)\varphi_j \right\|^2 = \sum_{j=1}^{D} (\hat{a}_j - a_j)^2.$$

Taking expectations and using independence of the X_i:

$$\begin{aligned}
\mathbb{E}\left[\|f_D - \hat{f}_D\|^2\right] &= \sum_{j=1}^{D} \mathbb{V}\mathrm{ar}(\hat{a}_j) = \sum_{j=1}^{D} \frac{1}{n} \mathbb{V}\mathrm{ar}(\varphi_j(X_1)) \\
&\leq \frac{1}{n} \sum_{j=1}^{D} \mathbb{E}[\varphi_j^2(X_1)] = \frac{1}{n} \mathbb{E}\left[\sum_{j=1}^{D} \varphi_j^2(X_1)\right] \\
&\leq \frac{1}{n} \left\| \sum_{j=1}^{D} \varphi_j^2 \right\|_\infty \leq \frac{\phi_0 D}{n}.
\end{aligned}$$

Variance bound: $\mathbb{E}\left[\|f_D - \hat{f}_D\|^2\right] \leq \dfrac{\phi_0 D}{n}$, where $\phi_0 = \left\|\sum_{j=1}^{D} \varphi_j^2\right\|_\infty / D$.

Risk Upper Bound

Combining the bias–variance decomposition with the variance bound:

Proposition 2.3.1 — Risk Bound

For the projection estimator $\hat{f}_D$ in an orthonormal basis (φ_j) satisfying $\| \sum_j \varphi_j^2 \|_\infty \leq \phi_0 D$:

$$\mathbb{E}\left[\| f - \hat{f}_D \|^2 \right] \leq \underbrace{\| f - f_D \|^2}_{\text{Bias}^2 \searrow D} + \underbrace{\frac{\phi_0 D}{n}}_{\text{Variance} \nearrow D} .$$

The two terms behave oppositely as D varies:

- Bias^2 decreases when D increases (S_D better approximates f).
- Variance increases when D increases (more coefficients to estimate).

$\Rightarrow$ An optimal D must balance these two terms.

Bias and Smoothness of f

The bias $\|f - f_D\|^2$ depends on how well S_D approximates f, which relates to the **regularity** of f.

Hölder Class

f belongs to the **Hölder class** of order α if:

$$|f(x) - f(x')| \leq C_f |x - x'|^\alpha, \qquad \forall\, x, x' \in [0,1].$$

Bias Bound for Histogram Basis

For f in the Hölder class with parameter α and the histogram basis with D bins:

$$\|f - f_D\|^2 \leq C_f^2\, D^{-2\alpha}.$$

Interpretation:

- Smoother f (larger α) $\Rightarrow$ bias decays faster with D.
- The same order $D^{-2\alpha}$ holds for trigonometric and other classical bases.

Bias Bound: Proof Sketch (Histogram Basis)

For the histogram basis, $f_D = \sum_{j=1}^{D} \langle f, \varphi_j \rangle \varphi_j$ and:

$$\|f - f_D\|^2 = \sum_{j=1}^{D} \int_{(j-1)/D}^{j/D} \left(f(x) - D \int_{(j-1)/D}^{j/D} f(u)\, du \right)^2 dx.$$

Within each bin $[(j-1)/D, j/D]$, both x and u satisfy $|x - u| \leq 1/D$. Using the Hölder condition $|f(x) - f(u)| \leq C_f |x - u|^\alpha$:

$$\|f - f_D\|^2 \leq C_f^2 \sum_{j=1}^{D} \int_{(j-1)/D}^{j/D} D^{-2\alpha}\, dx = C_f^2 D^{-2\alpha}.$$

Key message: The bias depends on the **regularity** α of f and the **bin width** $1/D$. A smooth f (large α) has small bias even with a coarse partition.

Optimal Dimension D^*

Combining the bias bound $\|f - f_D\|^2 \leq C_f^2 D^{-2\alpha}$ with the variance bound $\phi_0 D/n$:

The risk is bounded by:

$$\psi(D) := C_f^2 D^{-2\alpha} + \frac{\phi_0 D}{n}.$$

Minimizing $\psi(D)$ over $D > 0$:

$$\psi'(D) = -2\alpha C_f^2 D^{-2\alpha-1} + \frac{\phi_0}{n} = 0 \quad \Rightarrow \quad D_{\text{opt}} = \left(\frac{2\alpha C_f^2}{\phi_0}\right)^{\frac{1}{2\alpha+1}} n^{\frac{1}{2\alpha+1}}.$$

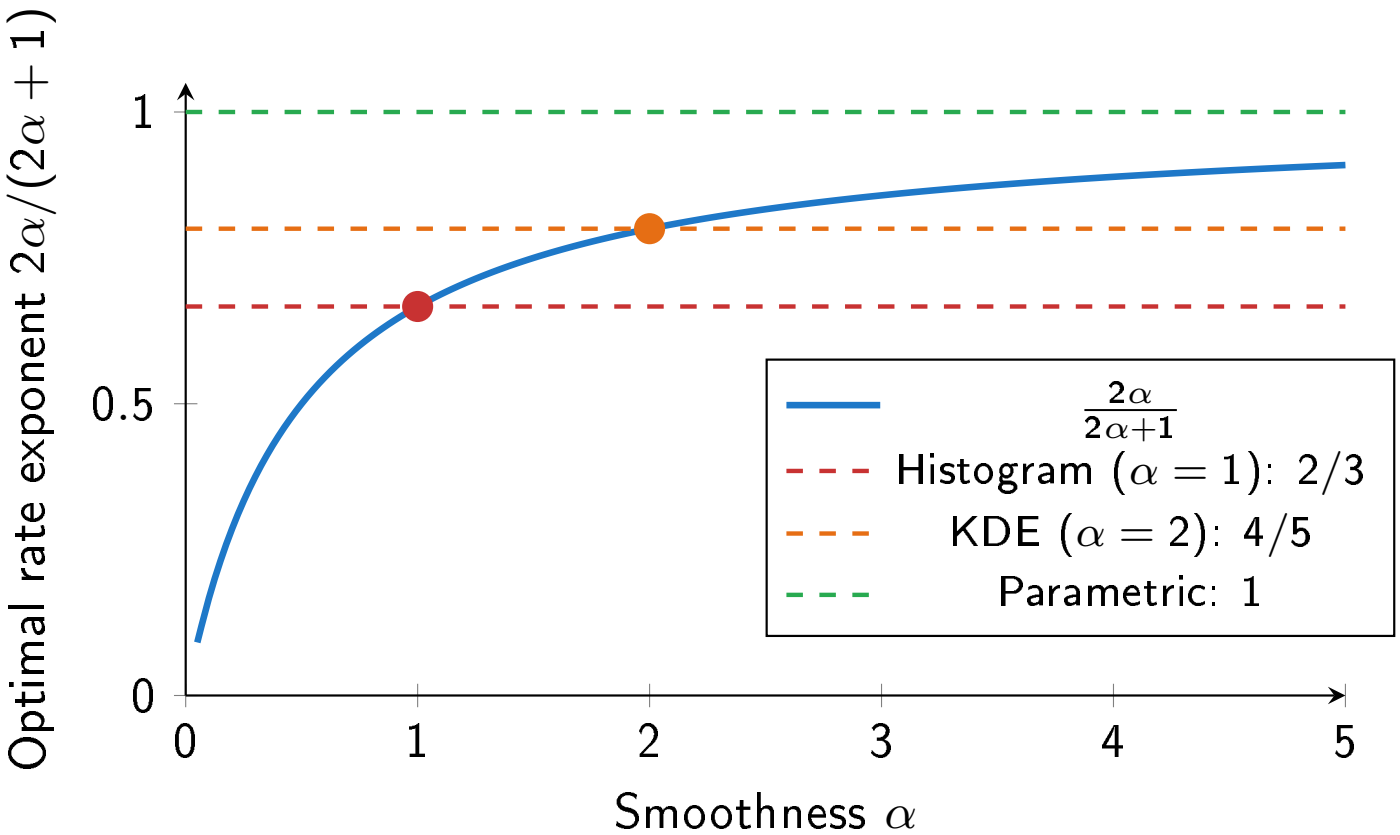
Optimal Rate

With $D = D_{\text{opt}}$:

$$\mathbb{E}\left[\|f - \hat{f}_{D_{\text{opt}}}\|^2\right] \leq C(C_f, \phi_0, \alpha)\, n^{-\frac{2\alpha}{2\alpha+1}}.$$

This is the **minimax optimal rate** for estimating $f \in \mathcal{H}(\alpha, C_f)$.

Convergence Rates: Summary



The smoother f is, the faster the rate converges toward the parametric rate n^{-1}.
Projection estimators achieve the optimal rate for *any* smoothness class.

The Oracle Dilemma

The optimal dimension is:

$$D_{\text{opt}} = \left(\frac{2\alpha C_f^2}{\phi_0} \right)^{\frac{1}{2\alpha+1}} n^{\frac{1}{2\alpha+1}}.$$

Problem: D_{opt} depends on:

- The smoothness α of f — unknown!
- The Hölder constant C_f — unknown!

$\Rightarrow$ D_{opt} is an oracle (cannot be computed from data alone).

Goal of adaptive estimation: Find a **data-driven** $\hat{D}$ such that $\hat{f}_{\hat{D}}$ performs as well as the oracle $\hat{f}_{D_{\text{opt}}}$, *without knowing* α *or* C_f.

The Bias–Variance Reformulation

We want to choose D to minimize:

$$\|f - \hat{f}_D\|^2 = \underbrace{\|f - f_D\|^2}_{\text{bias}^2} + \underbrace{\|f_D - \hat{f}_D\|^2}_{\text{variance}}.$$

Step 1: Simplify the bias term. By Pythagoras ($\langle f_D, f \rangle = \|f_D\|^2$):

$$\|f - f_D\|^2 = \|f\|^2 - \|f_D\|^2.$$

Since $\|f\|^2$ is a constant (independent of D), minimizing the total risk is equivalent to minimizing:

$$-\|f_D\|^2 + \|f_D - \hat{f}_D\|^2.$$

Step 2: Estimate both terms from data.

- Approximate the variance $\|f_D - \hat{f}_D\|^2$ by its expectation $\phi_0 D/n$.
- Estimate $-\|f_D\|^2$ by $-\|\hat{f}_D\|^2$ (note: $-\|\hat{f}_D\|^2 = \gamma_n(\hat{f}_D)$).

Data-Driven Dimension Selection

Penalized Model Selection

Define the **penalty** $\mathrm{pen}(D) = \kappa\phi_0 D/n$ and select:

$$\hat{D} = \mathrm{argmin}_{D\in\{1,\dots,n\}} \left\{ \gamma_n(\hat{f}_D) + \mathrm{pen}(D) \right\},$$

where $\gamma_n(\hat{f}_D) = \|\hat{f}_D\|^2 - \frac{2}{n}\sum_{i=1}^n \hat{f}_D(X_i)$ is the **empirical contrast**.

Key remarks:

- $\gamma_n(\hat{f}_D) = \min_{h\in S_D} \gamma_n(h)$ (verified earlier).
- $\kappa > 1$ is a numerical constant, calibrated by simulation.
- For density estimation, κ slightly above 1 works well in practice.
- The criterion $\gamma_n(\hat{f}_D) + \mathrm{pen}(D)$ is **computable from data only**.

For trigonometric basis: Since spaces are nested, the contrast $\gamma_n(\hat{f}_D)$ is computed **recursively** by adding one term at a time (efficient!).

The Oracle Inequality

Proposition 2.5.1 — Oracle Inequality

Assume nested models ($S_D \subset S_{D'}$ when $\dim(S_D) \leq \dim(S_{D'})$) and f bounded. There exist constants $C, C' > 0$ such that:

$$\mathbb{E}\left[\|\hat{f}_{\hat{D}} - f\|^2\right] \leq C \inf_{D \in \{1,\ldots,n\}} \left\{\|f - f_D\|^2 + \mathrm{pen}(D)\right\} + \frac{C'}{n}.$$

What this tells us:

- $\hat{f}_{\hat{D}}$ performs **as well as the best model** in $\{S_1, S_2, \ldots, S_n\}$, up to constant C.
- The remainder $C'/n = o(n^{-2\alpha/(2\alpha+1)})$ is negligible for large n.
- $\hat{f}_{\hat{D}}$ is an **adaptive estimator**: it automatically adjusts to the unknown smoothness α.

Consequence: $\hat{f}_{\hat{D}}$ achieves the rate $n^{-2\alpha/(2\alpha+1)}$ without knowing α.

Proof Sketch of the Oracle Inequality

Step 1: By definition of $\hat{D}$, for any D:

$$\gamma_n(\hat{f}_{\hat{D}}) + \mathrm{pen}(\hat{D}) \leq \gamma_n(f_D) + \mathrm{pen}(D). \qquad (*)$$

Step 2: A key algebraic identity for the contrast:

$$\gamma_n(t) - \gamma_n(s) = \|t - f\|^2 - \|s - f\|^2 + 2\nu_n(t - s),$$

where $\nu_n(t) = \frac{1}{n}\sum_{i=1}^n (t(X_i) - \langle t, f \rangle)$ is the **centered empirical process**.

Step 3: Combining (*) and this identity, and using $2xy \leq x^2/4 + 4y^2$:

$$\|\hat{f}_{\hat{D}} - f\|^2 \leq \|f_D - f\|^2 + \mathrm{pen}(D) + 4 \sup_{\|t\|=1} (\nu_n(t))^2 - \mathrm{pen}(\hat{D}).$$

Step 4: Control $\mathbb{E}[\sup_{\|t\|=1}(\nu_n(t))^2 - p(D, \hat{D})]_+$ using **Talagrand's inequality**.

This yields the oracle inequality after summing over all D'.

Example: $\mathcal{N}(4, 25)$ Density, $n = 1000$

Setup: $X_i \sim \mathcal{N}(4, 25)$ (mean 4, variance 25). True density: $f(x) = \frac{1}{5\sqrt{2\pi}} e^{-(x-4)^2/50}$.

Three estimators are compared on $[a, b] = [\min X_i, \max X_i]$:

- **Trigonometric, $D = 3$:** underfitting (too few terms), high bias.
- **Trigonometric, $D = 5$:** near-perfect reconstruction.
- **Histogram, $D = 10$:** good rough estimate, but discontinuous.

Key observation:
The trigonometric estimator with $D = 3$ is biased (too smooth), while $D = 5$ gives an almost perfect fit. The histogram with 10 bins gives a good overall shape but no smoothness.

For the trigonometric basis, increasing D by 2 only requires computing **two new terms**.

Adaptive Model Selection in Practice

Adaptive Procedure (Trigonometric Basis)

1. Normalize data: $U_i = (X_i - a)/(b - a) \in [0, 1]$.
2. For $j = 1, \ldots, d$ (with $d = \lfloor \sqrt{n} \rfloor$): compute $\hat{a}_{2j}, \hat{a}_{2j+1}$ and update $\gamma_n(\hat{f}_D)$ recursively.
3. Select $\hat{D} = \operatorname{argmin}_D \{\gamma_n(\hat{f}_D) + \kappa(2d + 1)/n\}$ with $\kappa \approx 0.25$.
4. Return $\hat{f}_{\hat{D}}$ restricted to $[a, b]$, rescaled by $1/\sqrt{b - a}$.

Results (typical):

- The adaptive selector finds a $\hat{D}$ that balances bias and variance automatically.
- For $\mathcal{N}(4, 25)$ with $n = 1000$, the selector typically chooses $\hat{D} \approx 5$.
- The estimate closely tracks the true density.

Advantage over histogram basis: Nested spaces allow recursive computation.
Disadvantage: The resulting estimate may be slightly negative in some regions.

Summary: Projection Estimator

Main Results

1. **Definition:** $\hat{f}_D = \sum_{j=1}^{D} \hat{a}_j \varphi_j$, $\hat{a}_j = \frac{1}{n} \sum_{i=1}^{n} \varphi_j(X_i)$.
2. **Risk:** $\mathbb{E}[\|f - \hat{f}_D\|^2] \leq \|f - f_D\|^2 + \phi_0 D/n$.
3. **Bias:** $\|f - f_D\|^2 \leq C_f^2 D^{-2\alpha}$ when $f \in \mathcal{H}(\alpha, C_f)$.
4. **Optimal D^*:** $D^* \propto n^{1/(2\alpha+1)}$, giving rate $n^{-2\alpha/(2\alpha+1)}$.
5. **Adaptive selection:** $\hat{D} = \operatorname{argmin}_D \{\gamma_n(\hat{f}_D) + \kappa \phi_0 D/n\}$ achieves the oracle rate without knowing α.

Projection vs. Kernel: Full Comparison

Property	**Projection Estimator**	**Kernel (KDE)**
Smoothing param.	Dimension D	Bandwidth h
Optimal rate	$n^{-2\alpha/(2\alpha+1)}$	$n^{-2\alpha/(2\alpha+1)}$
Adaptive selection	Penalized contrast	Cross-validation
Output	Series expansion	Smooth function
Always a density?	No (can be negative)	Yes (if $K \geq 0$)
Nested computation	Yes (trigon.) / No (hist.)	N/A
Theoretical tools	Oracle inequality	DKW, Talagrand
Basis flexibility	Many choices	Kernel choice

Both methods achieve the **same minimax optimal rate**. The projection estimator is particularly attractive when a specific function space structure is known.

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